

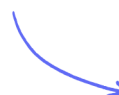
Structured Questions

Current, Potential Difference, Resistance & Power

Electric Current / Potential Difference / Ohm's Law / Charge Conservation in Circuits / Energy Conservation in Circuits / Resistance in Series & Parallel / Electrical Power / Current-Potential Difference Graphs

Medium (3 questions)	/24
Hard (2 questions)	/21
Total Marks	/45

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Medium Questions

- 1 (a) The photograph shows a statue of Buddha in Sri Lanka, which is protected by a lightning conductor.



© Valery Shanin/123RF

During a storm, a potential difference of 2.7 MV was generated between a cloud and the top of the lightning conductor on the statue. A flash of lightning passed between the cloud and the lightning conductor, producing a current of 25 kA for a time of 7.5 ms.

Calculate the energy transferred by the lightning strike.

Energy transferred =

.....

.....

.....

(3 marks)

- (b) The lightning conductor is a length of copper wire with a diameter of 1.2×10^{-2} m and a resistance of $4.3 \times 10^{-3} \Omega$. It runs along the back of the statue from the base to a height of 1.5 m above the top of the statue.

A guidebook claims that the statue is over 30 m high.

Assess the validity of this claim.

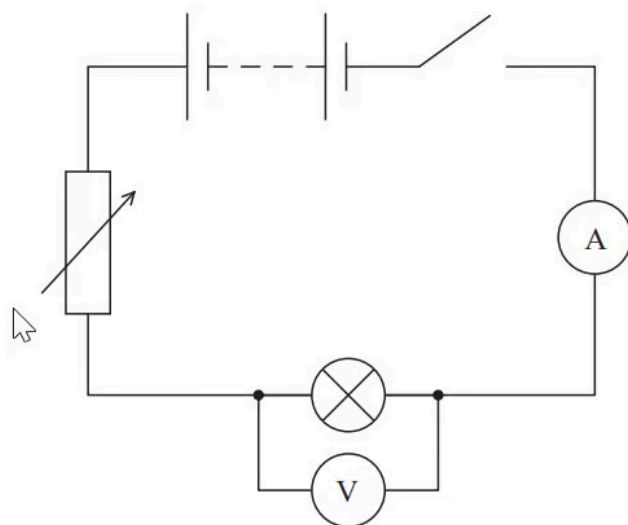
resistivity of copper = $1.7 \times 10^{-8} \Omega \text{ m}$

(4 marks)

- (c) Give a reason why the lightning conductor should be taller than the statue.

(1 mark)

- 2 (a)** A student set up the circuit shown and measured the current I through the filament lamp for a range of values of potential difference (p.d.) V .



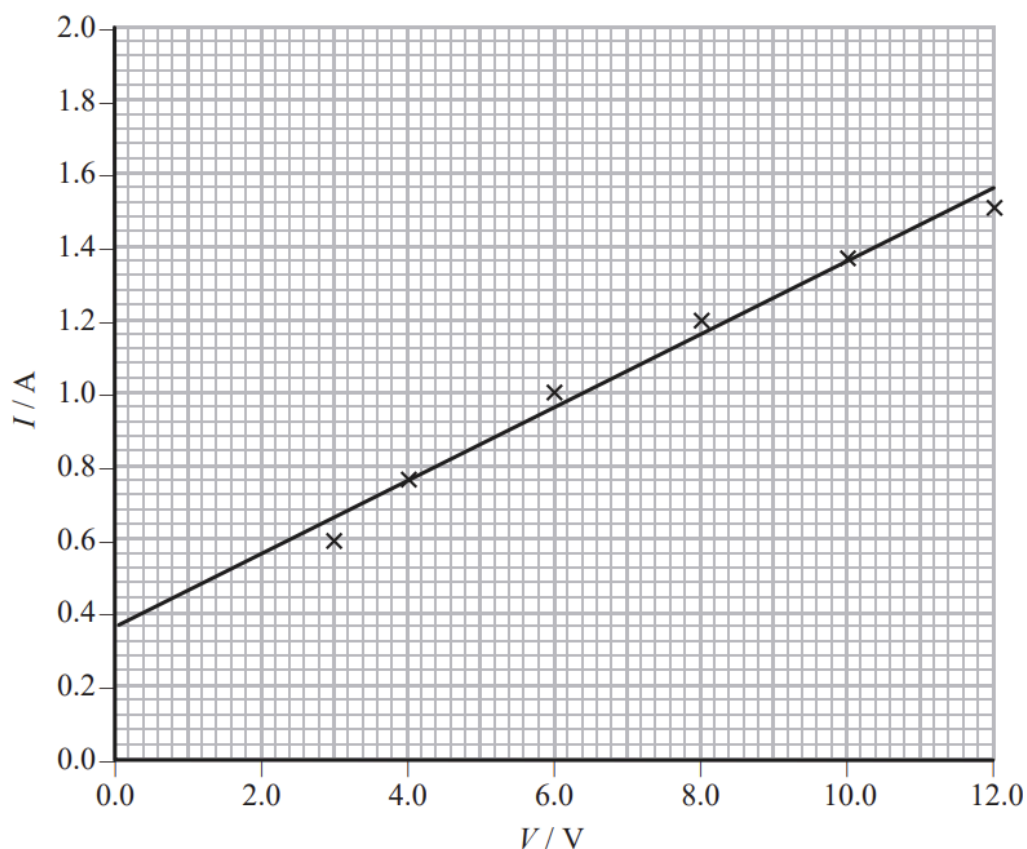
The student's data is shown in the table.

V / V	I / A
3.0	0.6
4.0	0.75
6.0	1.00
8.0	1.20
10.0	1.35
12.0	1.5

Criticise the student's recording of the data.

(1 mark)

- (b) The student drew a graph of how current varies with p.d. She drew a straight line on the graph and claimed that the data demonstrates that the filament lamp obeys Ohm's law because the graph is linear.



Assess the validity of the student's statement.

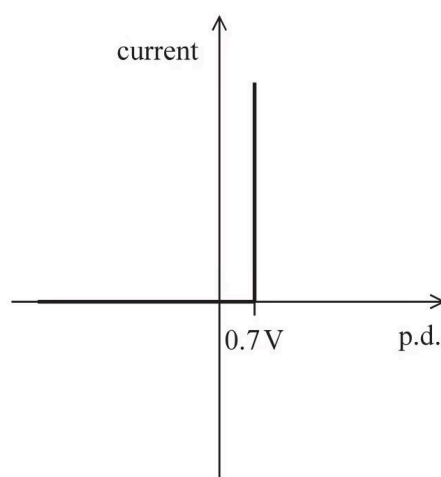
(4 marks)

- (c) Using the circuit shown the student was unable to obtain data for p.d.s less than 2.5 V.

Draw a diagram of a circuit the student could have used to enable a full range of p.d.s from 0 to 12 V to be investigated.

(2 marks)

3 (a) The graph shows how current varies with potential difference (p.d.) for an ideal diode.



Describe how the current through this diode varies for positive p.d.s and negative p.d.s.

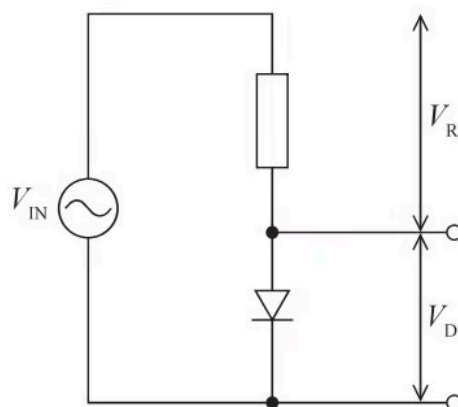
(2 marks)

(b) An alternating p.d. V_{IN} has a peak value of 3.4 V .

i) Calculate the r.m.s. value.

r.m.s. value =**(2)**

ii) V_{IN} is applied to a diode and resistor as shown.



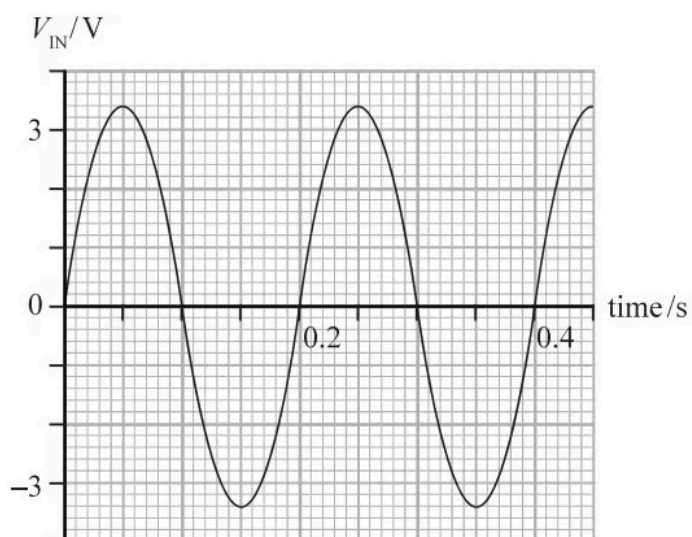
The p.d. across the resistor is V_R and the p.d. across the diode is V_D . V_D is the output.

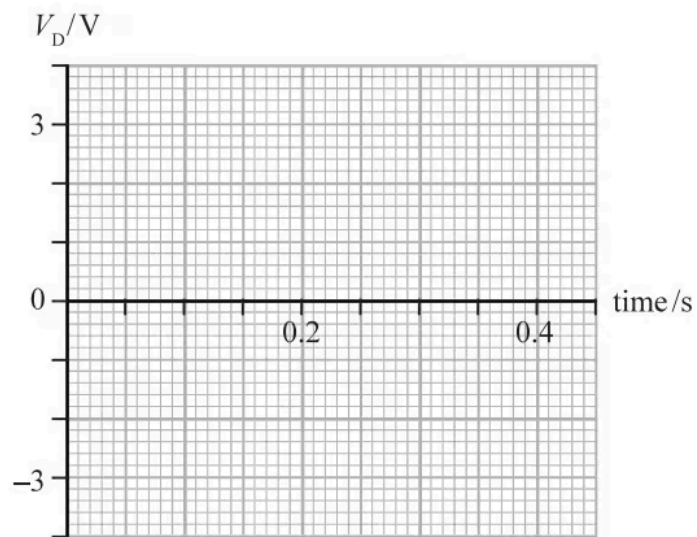
Explain why $V_{IN} = V_R + V_D$ at any given time.

(2)

iii) The graph shows how V_{IN} varies with time.

Sketch a graph of V_D against time using the axes provided below.





(3)

(7 marks)

Hard Questions

- 1 (a)** The world solar challenge is set every two years, in Australia. The challenge is to complete a three thousand kilometre route with a vehicle powered only by the Sun.

Vehicles have their surfaces fitted with solar panels, as shown in the photograph.



(Source: © LAURENT DOUEK/LOOK AT SCIENCES/SCIENCE PHOTO LIBRARY)

One of the solar panels has an e.m.f. of 8.2 V when in sunlight. The terminal potential difference is 5.5 V when a current of 0.45 A is drawn from the solar panel.

Calculate the internal resistance of the solar panel in these conditions.

Internal resistance =

.....

.....

.....

(3 marks)

(b) A bank of 380 of these solar panels is used to charge the battery in a vehicle.

The panels are connected in parallel and the current provided by each panel is 0.45 A.

When fully charged, the energy stored in the battery is 12 kWh.

Calculate the time, in hours, to fully charge this battery if the solar panels are in sunlight.
Assume the efficiency of charging this battery is 100%.

Time = hours

.....

.....

.....

(3 marks)

(c) The vehicle can reach a maximum speed of 34 ms^{-1} on flat ground. The electric motor used to move the vehicle has a power of 4.5 kW.

i) Calculate the initial acceleration of the vehicle as it starts from rest.

mass of vehicle and driver = 420 kg

Initial acceleration =**(3)**

ii) State one assumption made in this calculation.

(1)

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.....

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(4 marks)

- (d) Solar power alone would not be suitable for a family car because it is not sunny all the time.

Give two further reasons why solar power alone would not be suitable.

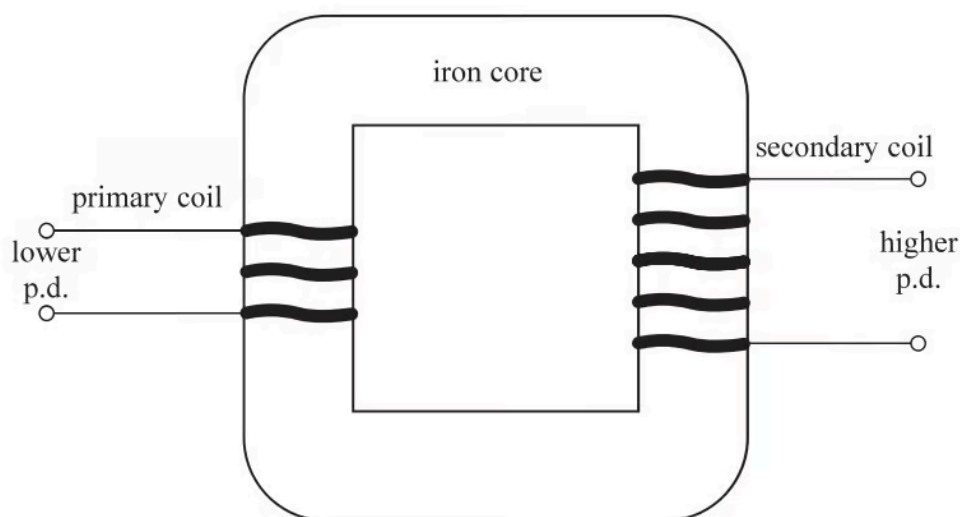
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..... (2 marks)

- 2 (a)** Electrical transmission systems are used to transmit electrical power from place to place.

Transformers are used to change potential differences (p.d.) and power transmission cables are used to transmit power.

The diagram shows a step-up transformer.



A step-up transformer is used to convert a lower p.d. to a higher p.d. An alternating p.d. is applied to the primary coil.

Explain how a higher p.d. is produced across the secondary coil.

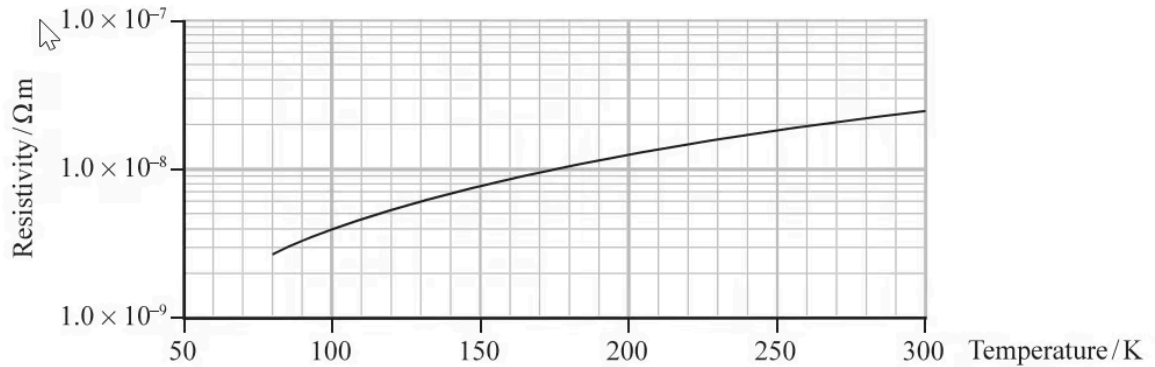
(4 marks)

- (b)** Efficient electrical transmission systems are being developed using superconductors.

Superconductors have zero resistance at low temperatures, and therefore no power is wasted by transfer to thermal energy unlike copper cable systems.

In one project a 1.05 km length of copper cable at a temperature of 270 K has been replaced by a superconductor. The superconductor has a cooling system which requires power.

The graph shows the variation of resistivity with temperature for copper.



Deduce whether the power requirement of the superconductor cooling system is less than the power losses in the copper cable.

transmission power = 40 MW

transmission potential difference = 110 kV

cross-sectional area of copper cable = 145 mm²

power requirement of cooling system for the superconductor = 7 kW

(5 marks)